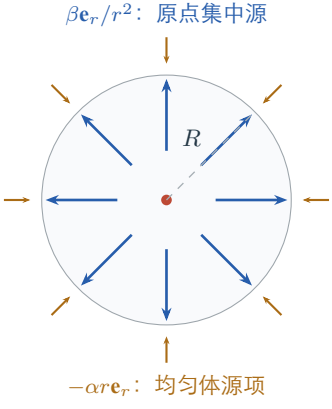
M1-73 径向场的体源与点源分别记账



$$E_r = -\alpha r + \frac{\beta}{r^2}$$
$$\Phi(R) = 4\pi(-\alpha R^3 + \beta)$$

$$\nabla \cdot \mathbf{E} = -3\alpha \quad (r > 0)$$
$$\nabla \times \mathbf{E} = 0$$

$$\nabla \cdot \mathbf{E} = -3\alpha + 4\pi\beta\delta^{(3)}(\mathbf{r})$$

箭头分别示意 $\alpha, \beta > 0$ 的两项。